By Ruth Sharon

Towards Monosemanticity: Decomposing Language Models With Dictionary learning

Why do we need interpretability?

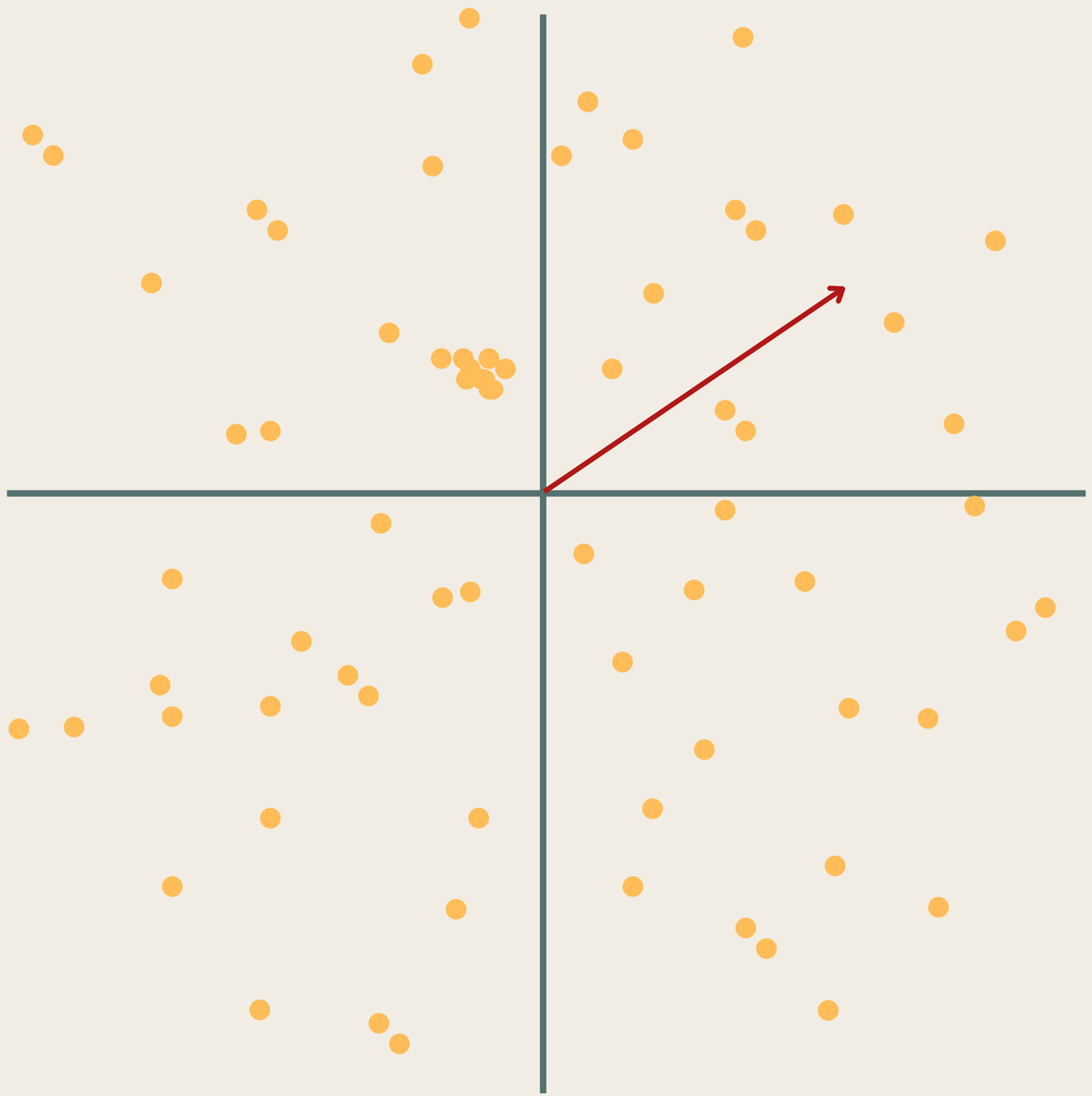
- A model's internal representations can be better understood by its individual components.
- Neurons are polysemantic, making them bad units of meaning.

"In some important way, we don't build neural networks. We grow them, we learn them."

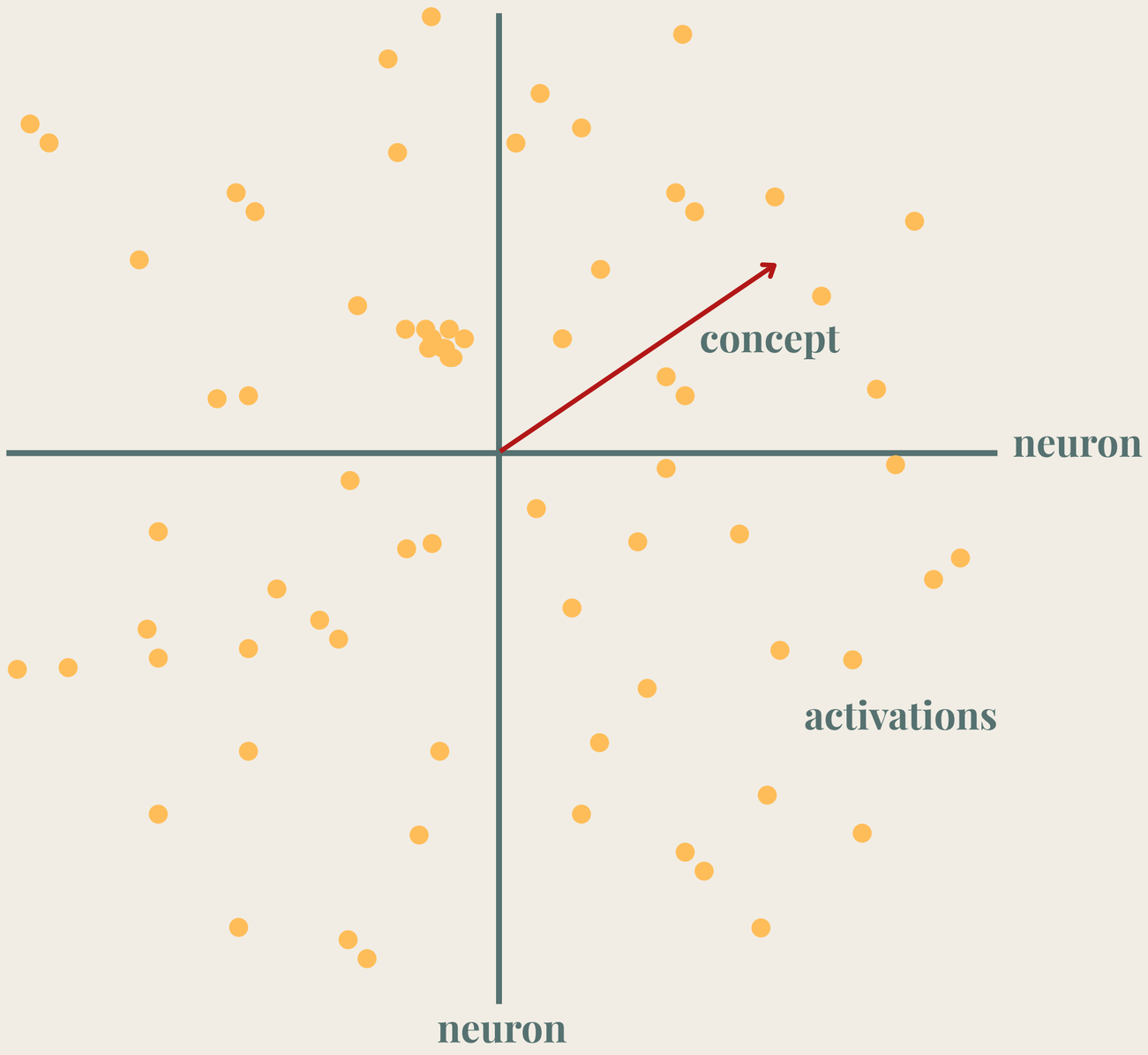
-Chris Olah, **Anthropic**

- Dictionary Learning/SAE learned directions in activation space
- Features allow cleaner and more stable interpretation

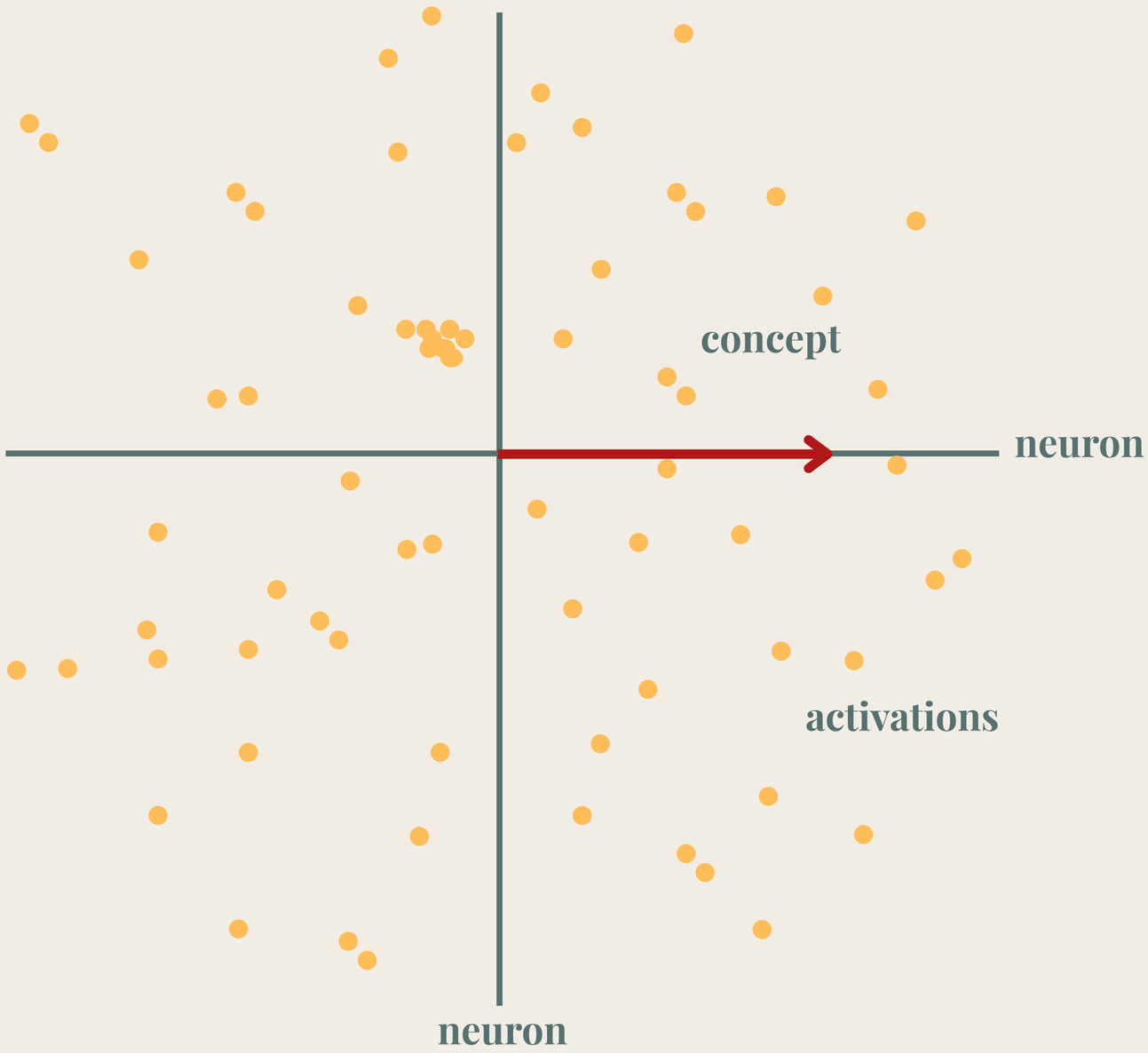
Problem Setup



Problem Setup



Problem Setup



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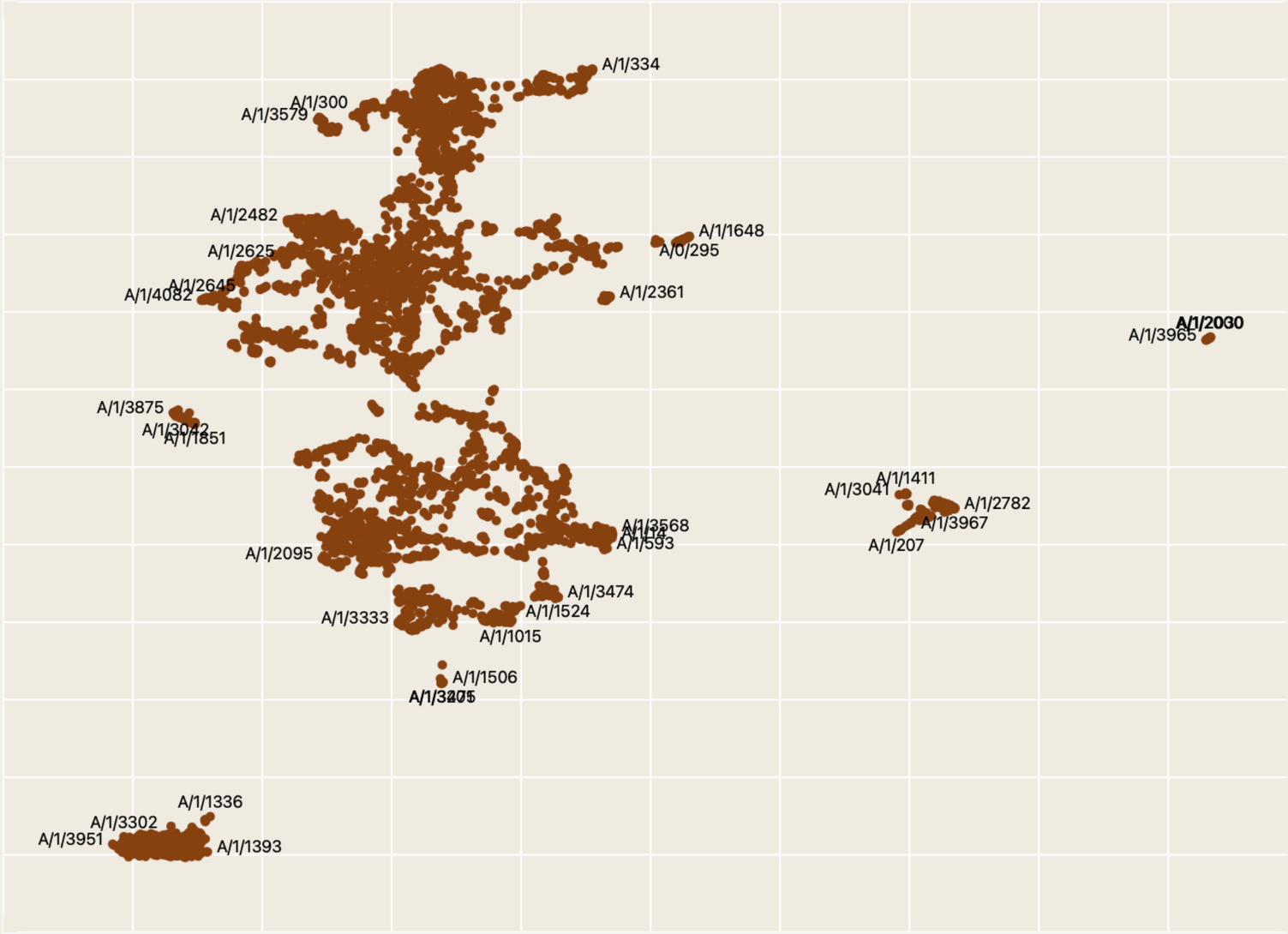


Interactive Exploration - A/1 and A/0 space of features

Problem Setup

Dataset Used

- Life 3.0 - Physics Heavy Chapters
- The Feynman Lectures on Physics - Volume I
- Quantum Mechanics Lectures - University of Oxford
- Black Hole Information Paradox - MIT Physics
- General Relativity, Black Holes, and Cosmology - University of Colorado

68164
words

87003
tokens

Approach

Extract Model Activations

Layer 8 of GPT2 | Feed corpus through model | Obtain Activation Vectors

Train a Sparse Autoencoder

Take activations as input

Reconstruct them using sparse, HD latent features

Identify and Interpret Features

Extract top tokens | Label features that show coherent concepts

Scales

Original Paper

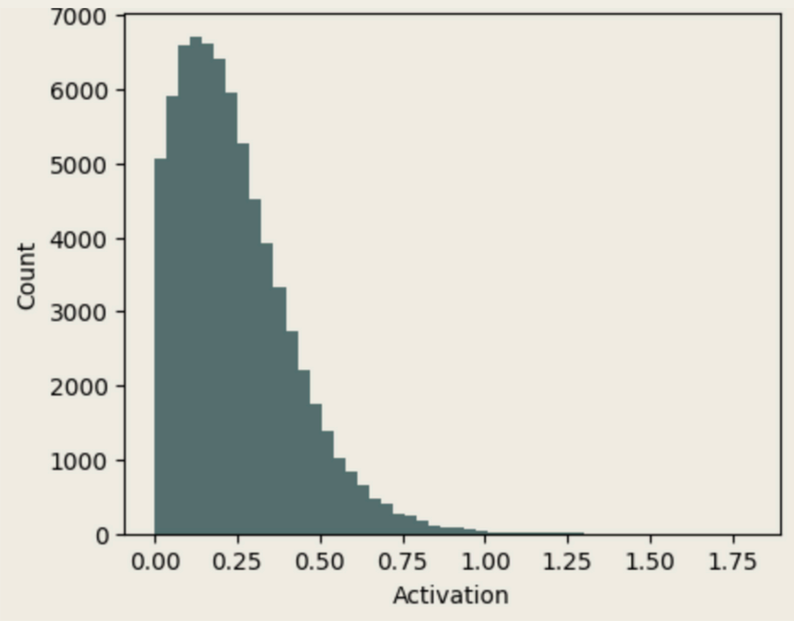
- One layer transformer with a 512-neuron MLP
- SAE trained on 8 billion activations
- Expansion factors from 1× (512 features) to 256× (131,072 features)
- Analysis on the 4096 learned features of the A/1 run

My Version

- Pre-trained GPT-2, analyzing layer 10 activations
- SAE trained on 180,233 token activations
- Expansion factor 8× (512 to 4096 features)
- Analysis on the strongest features

Results

FEATURE 8



MEAN ACT: 0.2049
MAX ACT: 1.8034

explosion until the cost of doing human-level work	1.017
the only matrices which do the trick, since any	0.881
so this does you no good if you	0.962
horizon area does not change	0.859
n(x2)} do not have to be orthogonally	0.902
Money didn't motivate Einstein exponentially just like you did, repeatedly	0.643
As gas falls downward toward the hole,	0.699
	0.788

OTHER STRONG FEATURES

transformed
converted
collapse
turned
into

will
should
can
might
may

catch up
know
sense
spirit

The bigger picture

- Despite being multimodal, a model's main mode of learning is text based.
- Models that understand the world through a text-driven lens must be understood through that same lens.
- As models grow more capable, monosemanticity allows us to see their inner workings and guide the concepts they learn.

Thank You!